University of Bahrain

Collage of Science

Department of Physics

Using steady state monte Carlo simulation to simulate light passing through tissue

by: Ali Isa Alsairafi

Student ID number: 20185402

Supervised by:

Dr. Habib Ashoor

This project reports is submitted in partial fulfillment of the

requirements for

the B.Sc. Degree at the Department of Physics

At the University of Bahrain

second semester

2021-2022

Abstract

Monte Carlo method is used to predict the probability of different outcomes in systems of random variables, in this thesis Monte Carlo method will be applied to simulate light propagation through tissue, by setting a number of light particles (photons) and giving each one of them a 'weight' , this weight will be reduced each time a photon gets scattered or absorbed, this weight is only a measure and is conventional, and we will be extended on in the theory section. The initial conditions are also big part of the simulation since we are going to compare light with different geometries such as collimated beam, gaussian beam, and square like beam, the main outcome of the simulation is that the collimated beam had the strongest fluence but with poor penetration, while the gaussian beam was the exact opposite starting the lowest fluence but it did not degrade as quickly as the collimated beam, and the square beam was somewhat in between leaning towards the gaussian beam in its behavior.

the simulation results are given as text files, that have the position of each photon and another text file that records the fluence rates in spherical, cylindrical, and planar coordinates. and the code used for the simulation is open source, so it will be included in the appendix section.

Contents

Abstract 2
Chapter 1: Introduction 4
Chapter 2: Theory (2) 5
2.1 Light properties 5
2.2 Photon step size 5
2.3 Photon's initial position on a circular plane 6
2.4 the fluence 7
2.5 Launch parameters 8
2.6 Threshold value 9
2.7 Launching photons 9
2.7.1 Hop 9
2.7.2 Drop 10
2.7.3 Spin 10
2.8 Terminate 13
2.9 Output 13
2.10 code structure 14
Chapter 3: Methodology 16
3.1 1: collimated launch 16
3.2 2: collimated Gaussian beam 16
3.3 3: plane like source 17
3.4 Tissue with no boundary conditions 18
3.5 Arm tissue 18
Chapter 4: Results and Analysis 20
4.1 Muscle tissue 20
4.2 Arm with multiple tissues 21
4.3 Comparison 23
4.4 Run time 24
Chapter 5: Discussion and Conclusion: 24
References 25
Appendix 26
5.1 Software used: 26
5.2 The code 26

5.3 graphing code 30
Acknowledgements..... 30

Chapter 1: Introduction

Monte carlo method was found by Von Neumann and Stanislaw Ulam to improve decision making in uncertain events (1), its named after Monte Carlo city located in Monaco, it has applications in many computational fields such as the Ising model to simulate magnetism, reducing the risk in the share market, and AI, cybersecurity etc.

The monte carlo method is used to predict the outcomes of uncertain events even if the event actual nature is deterministic.

In this thesis the monte carlo method is applied to simulate light passing through tissue, by launching a number of photons on the tissue and storing the value of the fluence until the photon loses all its energy, and when the energy of the photon reaches a certain limit the monte carlo method will take action to decide whether the photon will get to live for another cycle or be terminated.

The importance of this simulation is that light has a lot of applications in medical physics, applications such as x ray spectroscopy, tattoo removal, cancer treatment, hear removing and the list goes on and on.

By simulating light in this manner, one can predict how effective is a certain light beam at a certain task.

Chapter 2: Theory (2)

2.1 Light properties

When light passes through an object, multiple phenomena happen for the sake of this simulation we will only consider the elastic scattering, and absorption.

These two phenomena are bound by constants μ_a absorption coefficient, and μ_s scattering coefficient, now to be precise about their definition, the scattering coefficient μ_s can be described as the probability of scatter per infinitesimal unit length, and the absorption coefficient μ_a is the probability of absorption per infinitesimal unit length, or to put it simply these two parameters are a measure of how much a material can scatter or absorb light. these coefficients change as the incident wavelength changes, this is due to the fact that different lights with different energies react differently with matter.

Now let us talk about the weight of each photon, each photon is given a weight this is not the same weight as one might expect classically since photons are massless particles, the weight given to each photon is just a notation for energy of the photon for convention we will assume each photon starts with weight w=1, and each time the photon reacts with the matter

$$w = w_0(1 - albedo) \tag{1}$$

Where:

w_0 is the weight before interaction

$albedo$ is a quantity equal to $albedo = \frac{\mu_s}{\mu_s + \mu_a}$

2.2 Photon step size

Now after talking about the properties of light, we should talk about the trajectory of the incident light, each photon will be set in an initial place with an initial speed in a certain direction, in the experimental section we will be talking about these initial conditions, after giving a photon an initial condition and speed, we should talk about the step size, the step size should be related to the scattering and absorption coefficients the normalized probability density (2) is equal to:

$$P(s) = \frac{e^{-\mu_t s}}{\mu_t} \tag{2}$$

Where:

$\mu_t = \mu_s + \mu_a$

And s is the step size

Now we need to integrate P(s) and solve for s

$$\int p(s)ds = \int_0^{\infty} \frac{e^{-\mu_t s}}{\mu_t} = 1$$

This is gaussian distribution integral, without going into mathematical details the solution is equal to

$$\int p(s)ds = 1 - e^{-(\mu_t s)}$$

Now since this integral is a distribution, we should use the monte Carlo sampling (3)

$$rand = 1 - e^{\mu_t s}$$

Now we solve for s

$$s = -\frac{\ln(1 - rand)}{\mu_t}$$

Since we are dealing with probabilities these two quantities are equivalent $1 - rand \equiv rand$

And the step size is equal to

$$s = \frac{-\ln(rand)}{\mu_t} \tag{3}$$

This is how the step size is determined each time a photon reacts with the object.

2.3 Photon's initial position on a circular plane

In this simulation we will assume the light source is circular plane and particles are fired from inside a circle with a radius 'a'

$$p(r) = \frac{2\pi r}{\pi a^2} = \frac{2r}{a^2} \tag{4}$$

The assumption here is that each distance from the center of the circle has a probability of all points that are the same distance from the source and is the equivalent of a ring.

So now let us normalize and integrate the probability density:

$$A\int_0^a p(r)dr = A\int_0^a \frac{2r}{a^2}dr = 1 \tag{5}$$

The solution is

$$\frac{r^2}{a^2} = \frac{1}{A}$$

To account for the randomness of the process we change r and A so that they are random

$$\frac{r_r^2}{a^2} = \frac{1}{A}$$

And since r is random, we can make 1/A a random number for the computer to understand and we finally get

$$r_r = a\sqrt{rand} \tag{6}$$

2.4 the fluence

in this simulation we will track the flux and fluence of the incident light, with the fluence rate

$$\phi\left[{}^{W}/_{cm^2}\right] = P\,T(r) \tag{7}$$

And the fluence

$$\psi = Q\,T(r) \tag{8}$$

Where

P is the power and Q is the delivered energy

2.5 Launch parameters

Let us discuss the launch parameters that will affect each photon as it propagates through the tissue, these parameters are:

μ_a $[cm^{-1}]$ absorption coefficient

μ_s $[cm^{-1}]$ scattering coefficient

g $[-]$ anisotropy of scattering of tissue

$n[-]$ refractive index

[-] means dimensionless

These parameters are for an infinite tissue, but for boundaries we will change these parameters depending on the type of tissue, or its volume or and so on. More about the boundaries in the methodology section.

After specifying the boundaries of the tissue we are interested in, we need to do what is called binning, which simply means dividing the tissue into smaller pieces and measuring the parameters in each one of these pieces, this process will give us a clearer idea about the distribution of photons and save valuable time in the running process, and these are the binning parameters.

$N_z[-]$ is the number of bins in cartesian coordinates

$N_r[-]$ is the number of bins in spherical or cylindrical coordinates

$dz[cm]$ size of each bin in cartesian coordinates

$dr[cm]$ size of each bin in spherical or cylindrical coordinates

Where $dz = \frac{D}{N_z}$

And the final parameter is the number of photons

$N_p[-]$

2.6 Threshold value

Each photon is assigned a wight of 1 in the biggening and each time it reacts with the tissue the weight drops by a fraction of the current weight as in equation (1), so in order to annihilate photons we need to specify a value that if the photon reaches this value it will be annihilated, we will call this value the annihilation (ANH)

$$ANH = albedo^{N_s} \tag{9}$$

Solving for the number of steps

$$N_s = \frac{\ln(ANH)}{\ln(albedo)} \tag{10}$$

2.7 Launching photons

Now let us talk about the about the initial launch trajectories of each photon, the trajectory will be specified using the cosines (ux,uy,uz), but since the signal to noise in cartesian coordinates is much higher that it is in spherical or cylindrical coordinates because there are less parameters to fill we are going to convert the signal to those coordinates while also keeping the signal in cartesian coordinates.

The trajectories are

$$ux = \cos(\theta)\cos(\phi) \tag{11}$$

$$uy = \cos(\theta)\sin(\phi) \tag{12}$$

$$uz = \cos(\theta) \tag{13}$$

We will test few trajectories that will be talked about in the methodology section.

2.7.1 Hop

Earlier we determined the step size of the photon, after specifying its trajectory we need to update its position with the relation with the step size from equation 3

$$x_{n+1} = x_n + s \times ux \tag{14}$$

$$y_{n+1} = y_n + s \times uy \tag{15}$$

$$z_{n+1} = z_n + s \times uz \tag{16}$$

Until the photon gets terminated.

2.7.2 Drop

Upon interacting with the tissue the photon is absorbed and scattered, after the photon's absorption it drops into a bin and its weight drops by an amount $\frac{\mu_a}{\mu_t}$

Which keeps the weight

$$w = w(\frac{\mu_s}{\mu_t})$$

The photon needs to be dropped into a bin, and since the number of bins is an integer we have to round its position to be also an integer so the radial bins will look like

$$r = \sqrt{x^2 + y^2}$$

$$ir = floor\left(\frac{r}{dr}\right) + 1 \tag{17}$$

And for the depth

$$iz = floor\left(\frac{z}{dz}\right) + 1 \tag{18}$$

The floor function is a function that takes any double and round it to the integer below it, 1 is added to the function just to ensure we do not have a zeroth bin.

After this event the photon's energy is absorbed and what remains is the scattering event.

2.7.3 Spin

After the photon's weight drops due to absorption, its trajectory should also change due to scattering.

To update the photon's trajectory we need to specify two angles the azimuthal and deflecting angles φ and θ.

To find the deflecting angle θ we will use Henyey-Greenstein function, which was proposed in 1941, similar to what we did before let us start with probability density proposed by Henyey-Greenstein, expressed in terms of the deflecting angle θ and the anisotropy of scattering g:

$$p(\cos\theta) = \frac{1}{4\pi}\frac{1-g^2}{\left(1+g^2-2gcos(\theta)\right)^{\frac{3}{2}}} \tag{19}$$

Since it is a probability density its integration must be equal to one

$$2\pi\int_0^{\pi} p(\cos\theta)sin\theta d\theta = 1 \tag{20a}$$

And the definition of g is

$$2\pi\int_0^{\pi} p(\cos\theta)\cos(\theta)\sin(\theta)\,d\theta = g \tag{20b}$$

As we did before we integrate and assign a random number for the monte carlo sampling

But this time it is a system of equations instead

If g=0	If g=1	Otherwise
$cos(\theta) = 2RND - 1$	$cos(\theta) = 1.0$	$cos(\theta) = \dfrac{1+\mathrm{g}^2 - \left(\dfrac{1-\mathrm{g}^2}{(1-\mathrm{g}+2\mathrm{g}\times\mathrm{RND})^{\frac{3}{2}}}\right)^2}{2g}$

as for the azimuthal angle things are less complicated where we just need a random number from the interval $(\mathbf{0}, \mathbf{2\pi})$

$$\varphi = RND \times 2\pi \tag{21}$$

Now using the angles, we found we will update the trajectory as follows

$$\sin(\theta) = \sqrt{1-\cos^2\theta}$$

$$\text{temp} = \sqrt{1 - \text{uz}^2}$$

temp is a dummy variable

$$\boldsymbol{u}\text{xx} = \frac{\sin\theta(\text{ux uz}\cos\varphi - \text{uy}\sin\varphi)}{\text{temp}} + \text{ux}\cos\theta \tag{22}$$

$$\text{uyy} = \frac{\sin\theta(\text{uy uz}\cos\varphi - \text{ux}\sin\varphi)}{\text{temp}} + \text{uy}\cos\theta \tag{23}$$

$$\text{uzz} = -\sin\theta\cos\varphi\,\text{temp} + \text{uz}\cos\theta \tag{24}$$

In the special case of the trajectory being so close to the z axis $\boldsymbol{uz} \approx \pm\mathbf{1}$ we will use another set of equations

$$uxx = sin\theta\ cos\varphi$$

$$uyy = sin\theta\ sin\varphi$$

If $\boldsymbol{uz} \geq \mathbf{0}$ then

$$uzz = cos\theta$$

Else

$$uzz = -cos\theta$$

And finally, we update the trajectories into

$$ux = uxx$$

$$uy = uyy$$

$$uz = uzz$$

2.8 Terminate

After updating the trajectory and the photon's lost most of its weight, it is time for it to be gone but to preserve the energy of the system the photon will only be terminated once every ten times its weight goes below the threshold value.

2.9 Output

Now that all photons have been sorted into arrays of each coordinate system, we need to find the photon/bin value, in order to do that we just need to find the volume of each bin in cylindrical coordinates

$$V = 2\pi(ir - 0.5)dr^2dz \tag{25}$$

And in planar coordinates

$$V = dx\ dy\ dz \tag{26}$$

So, the value of photon weight per bin

$$A(i) = \frac{A(i)}{V(i) \times N_{photo}} \tag{27}$$

A(i) and V(i) are generalized for any coordinate system

And now for the value we are interested in which is the transmittance we simply divide A by the absorption coefficient

$$T = \frac{A}{\mu_a} \tag{28}$$

This value will be recorded in a file after compiling the code, it is also worth noting that the last position of each photon will be recorded in a separate file.

2.10 code structure

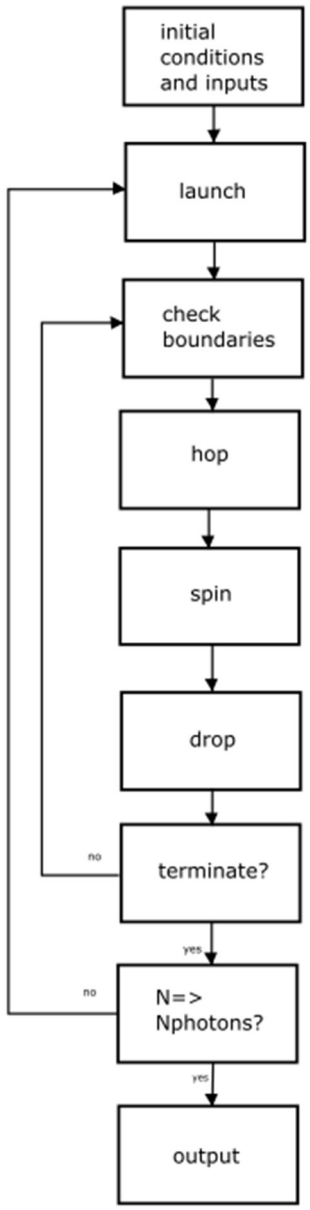


Figure 1 code structure

The code will start by assigning all the important parameters like the light properties and the position, then launch photons one by one, in each photon it will go through all the phases and reactions

While reducing its weight, when the photon reaches a certain weight the code will ask if it should terminate the photon or not, if the photon got terminated it will ask another question whether it should launch another photon or not, if the number of photons = total number of photons then the code will produce the output as a text file.

Chapter 3: Methodology

In this section we will talk about the tissues we will test the simulation on as well as the beam geometries and position.

The beam geometry:

1- Collimated launch
2- Gaussian beam
3- **Line like source**

3.1 1: collimated launch

This type of beam is straight forward the trajectory of the beam as it leaves the source is as follows

$$\boldsymbol{x = 0, ux = 0}$$

$$\boldsymbol{y = 0, uy = 0}$$

$$\boldsymbol{z = 0, uz = 1}$$

The photon will leave from the center and move towards the z axis

3.2 2: collimated Gaussian beam

This type of beam is a little more complicated where we need to specify a probability density

For the initial position of the beam such that:

$$p(r) = \frac{e^{\left(-\frac{r}{b}\right)^2} 2\pi r}{\pi b}$$

as we did before we are going to integrate the equation and use the monte carlo sampling method to get

$$\boldsymbol{r = b\sqrt{-ln(rand)}}$$

My choice of b will be $b = \frac{\pi}{2}$

Now the beam's initial condition will be

$$\boldsymbol{x = b\sqrt{-ln(rand)}}$$

$$\boldsymbol{y = 0}$$

$$\boldsymbol{z = 0}$$

The beam will span the x axis in this situation, and it is possible for us to chose the y axis instead if it is more compatible with the sample at hand.

As for the initial trajectory of the beam it will be the same as before

$$\boldsymbol{ux = 0}$$

$$\boldsymbol{uy = 0}$$

$$\boldsymbol{uz = 1}$$

3.3 3: plane like source

Perhaps one is interested in seeing how a square light source will affect the simulating

And to get that we need to make the photon start from -1 to 1 in both the x and y axis

$$\boldsymbol{x = sign(rand(), 2.0 * rand() - 1)}$$

$$\boldsymbol{y = sign(rand(), 2.0 * rand() - 1)}$$

$$\boldsymbol{z = 0}$$

As for the trajectory

$$\boldsymbol{x = 0}$$

$$\boldsymbol{y = 0}$$

$$\boldsymbol{z = 1}$$

We will assume that the light has wavelength of around 600 nm since the optical properties as mentioned before change with wavelength, also the number of photons fired will be 5000 for the muscle tissue and 10000 for the arm example.

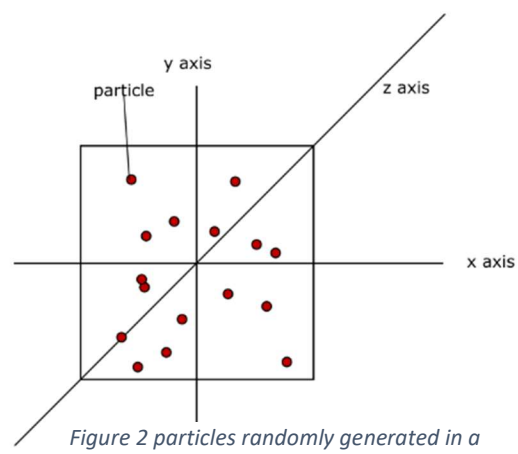


Figure 2 particles randomly generated in a square lattice

Now let us move on into the tissue

3.4 Tissue with no boundary conditions

The optical properties of the muscle are

$$\mu_a = 0.2\ \mathrm{cm}^{-1}$$

$$\mu_s = 255.41\ \mathrm{cm}^{-1}$$

$$g = 0.72$$

3.5 Arm tissue

As for the second type of tissue we will assume that the light is put upon an arm with many different tissues, in this example I used my arm and found its circumference to be = 28cm

And by dividing by 2π we get my arm's radius. $r_{hand} = 4.46\ cm$

Then I used a picture of a cross section of an arm and scaled it with my arm to approximate the length of each type of tissue.

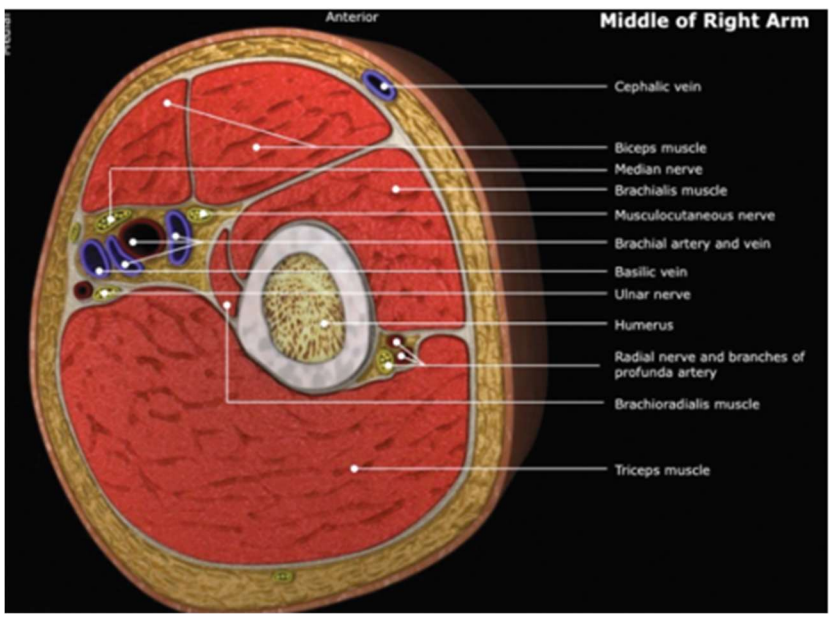


Figure 3: cross section of an arm, (4)

Given the knowledge and time during the simulation I found out that this approximation is good enough but since it is a simulation there is always room to improve

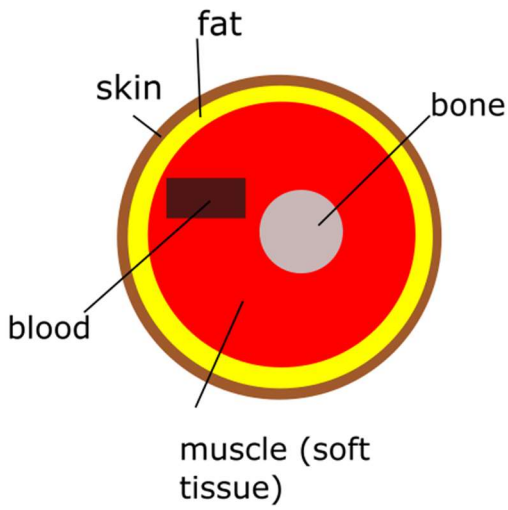


Figure 4: approximation and scaling using Inkscape

This table shows each type of tissue's optical properties and the boundary conditions in the simulation.

Table 1 light properties of certain tissues

Tissue type	μ_s [cm]	μ_a [cm]	Source
Skin	**268.01**	**11.0**	(5), (6)
Fat	**157.03**	**0.15**	(7)
Muscle	**255.41**	**0.2**	(6)
Bones	**193.4**	**20.0**	(6)
Blood	**188.3**	**11.0**	(6) (8)
Air (albedo=0)	**0**	**0**	**--**

In this example the initial conditions are no longer initial but are dynamic and depend on the photon's position, it was also assumed that there are no conditions on the x axis since the arm is relatively long.

The anisotropy of all tissues was assumed to be g=0.9, because at wavelengths higher than 600 nm the anisotropy seems to not deviate a lot from 0.9 and because it is hard to directly measure the anisotropy.

Chapter 4: Results and Analysis

After successfully compiling the code, it is time to talk about the results.

4.1 Muscle tissue

The output for when there are no boundary conditions on a muscle tissue are as follows:

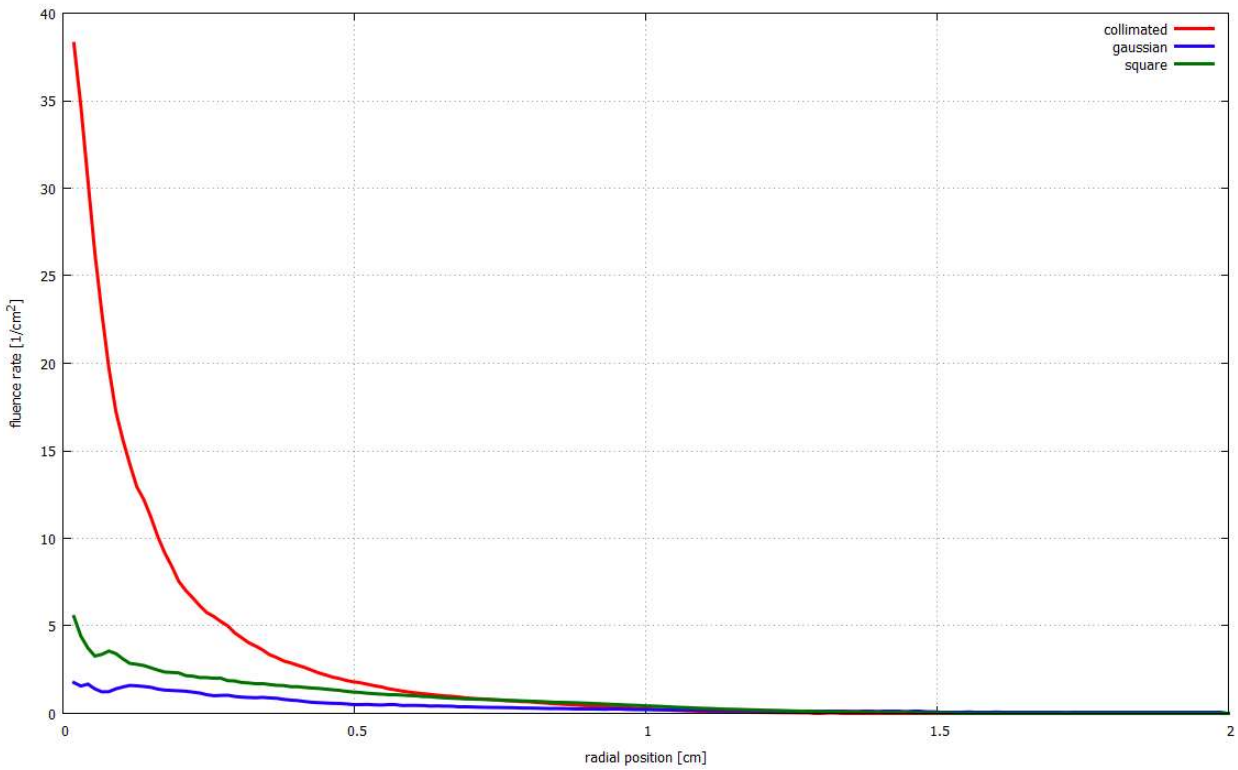


Figure 5 differences between 3 beams

As we can see the collimated beam had a higher fluence at first, but to see an in-depth comparison we will combine all pictures in one picture and change the scale of the y axis to log scale with base 10

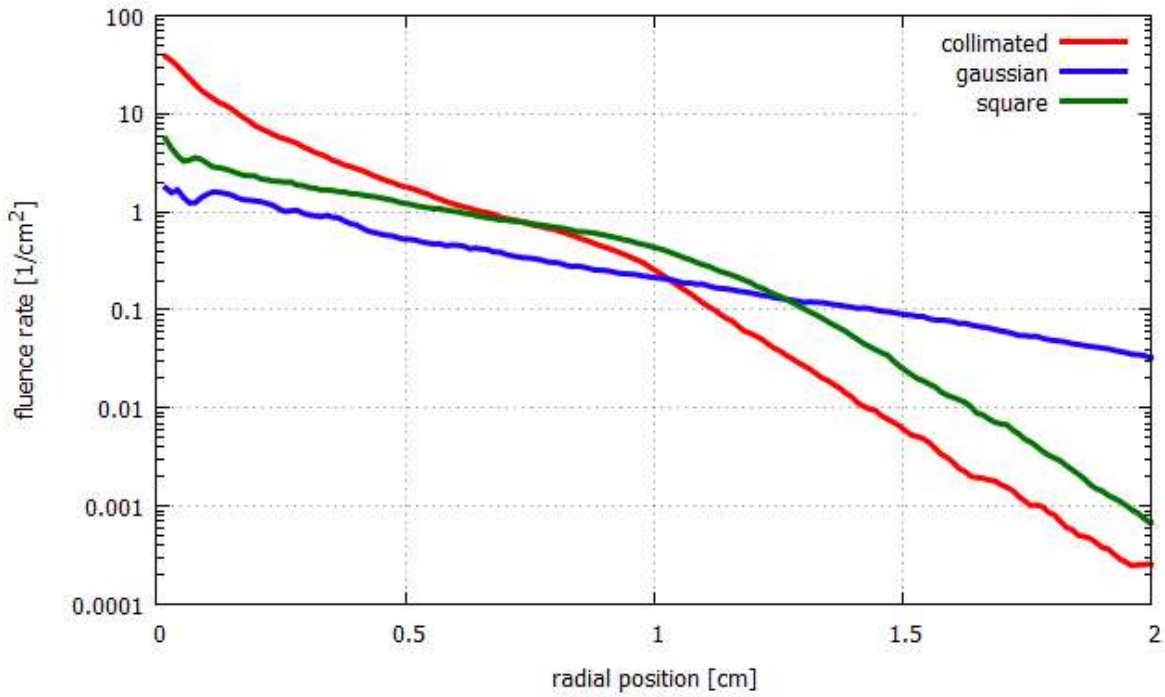


Figure 6 comparison between beams

Things now are more clear and precise, we can see that even though the collimated beam had initially a stronger fluence it quickly falls off as we get deeper into the tissue, while the other two beams are having more steady fall, one more thing to note is that the gaussian beam has another a second peak at around 0.2cm, if we can somehow tune a certain parameter in the beam we can perhaps use it to light a specific area inside a tissue.

4.2 Arm with multiple tissues

As we did before, let us start **by showing** the results

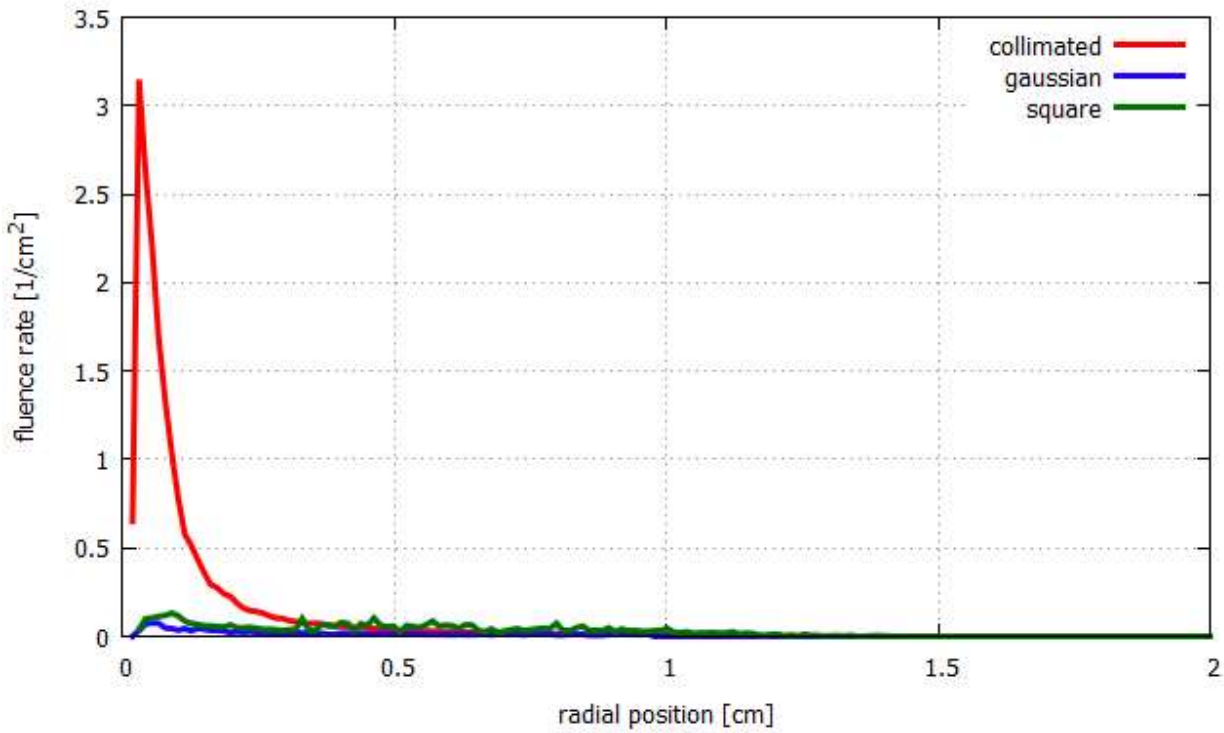


Figure 10 square beam, arm

There are clear trends that differ from the previous example, but now let us first use logscale to get a clearer idea

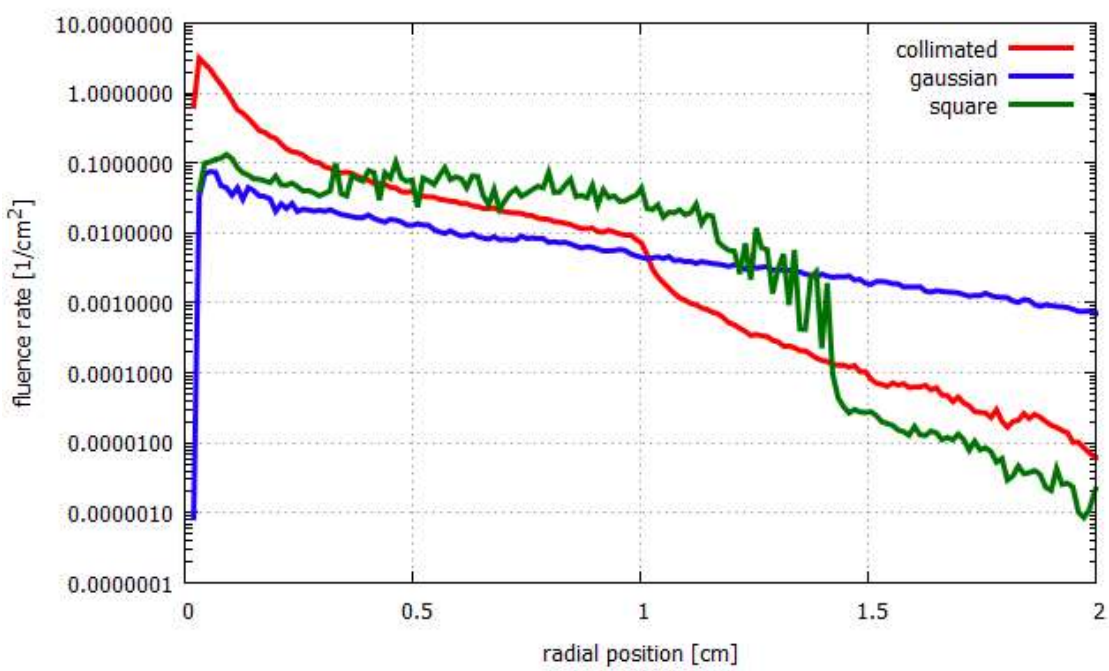
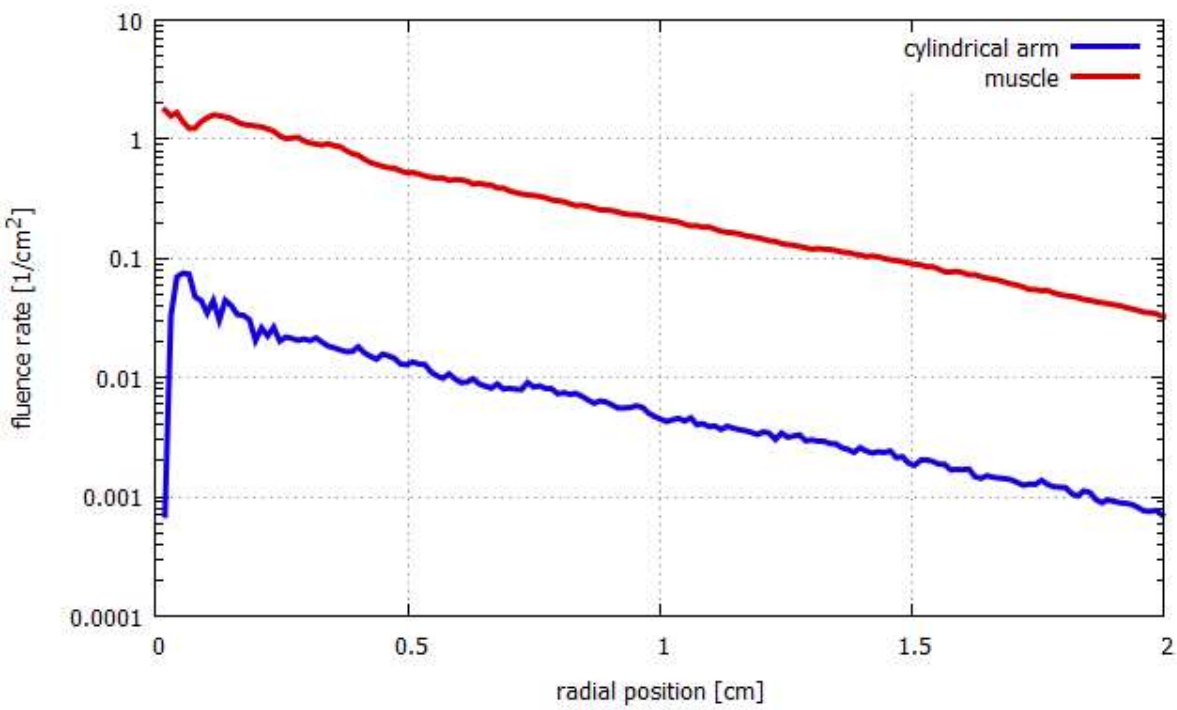


Figure 6 comparison between all beams

The gaussian beam showed a steady decline even thought it started from a very low point while the trend with the collimated beam having the fastest decline is still apparent, the square beam had a very fluctuated behavior, this is due to it interacting with more tissue types as it penetrates the arm. And one thing to note is that all beams had a local peak at around r=0.1 where the fat tissue which has less scattering and absorption coefficients when compared to skin tissue.

4.3 Comparison

Now to get an even more precise idea we need to compare the tissue with boundary conditions and the tissue without any limitations, this will serve as indicator for if the boundary conditions are actually affecting the output or not, we will use the gaussian beam in this comparison since it had the most difference between the two options.



On the surface this might not seem like a big difference but looking closely the fluence in the muscle is a lot higher than it is the arm tissue, even though the number of photons fired at the arm is double the number fired at the no boundary muscle, which makes sense considering that the light has to pass through the skin which has a very high absorbance and scattering coefficients which in turn means that the light will not be able to propagate through the tissue as in the “muscle-only” example.

One more thing to note is that in the first few millimeters there is a peak in the graph for arm, this peak corresponds to a well in the muscle example, this is a clear indicator that the beam

entered a different tissue (the fat surface) at around 0.1cm which was put in the boundary conditions.

4.4 Run time

The simulation takes about 6-7 seconds for 10000 photons, you can use this information to estimate the run time for a different number of photons, though it is necessary to point out the hardware used to get these results, the most important component for simulations like this is the cpu and the cpu used for this one is [AMD Ryzen 5 3600 6-core processor], the results will vary depending on the hardware used so there is no clear formula for the run time that will work on all devices.

Chapter 5: Discussion and Conclusion:

The simulation went smoothly and the only error that needs to be pointed out is that I had to start the beam from inside the tissue because the array that stores the photons position is not defined to accept negative values, this could be solved by either letting it accept negative values, or force the photons position to be zero whenever it becomes negative but that will break the physics of the simulation so the solution used is to make the beam start inside the tissue. The problem itself happens because of when photons are scattered there is a slight chance that they will go below the zero point. And now I would like to take this opportunity to talk about things that could improve the simulation even further,

1- first of all one might consider the time dependency of the simulation this is not a very complicated task you need to divide the step size by the speed of light and store the result in a bin so that you do not have make a time slot for every photon individually at every step it takes, and the bin size should be small enough so that results will be as close as possible to the real time of each photon.
2- One might think of introducing other properties to measure like the brightness of the light at each point.
3- Test the simulation on other types of tissues or even non biological materials like glass or water.
4- Try different light types or even polychromatic light, since the light properties change with the wavelength, we can give each photon a different wavelength and change the light properties of the material depending on which photon it is reacting with

5- Last suggestion is to do the reverse simulation to find the light properties of a said tissue.
6- Cube pins, this simulation deals with the bins as planes be it spherical or planar, but by making the bins into cubes or small spheres the first thing that will happen is that the definition of fluence needs to be changed for the new bin type, as for the second important thing is that we will trade the run fast run time of this simulation with the resolution of the results because we will get a clear idea of where in the tissue we are measuring.

These are mere suggestions that I hope people could work on in the future and come up with their own ideas since the sea of computational physics is huge and there will always be room to improve the current models.

References

1. **IBM cloud learn hub . [Online] IBM. https://www.ibm.com/cloud/learn/monte-carlo-simulation#:~:text=The%20Monte%20Carlo%20Method%20was,to%20a%20game%20of%20roulette..**

2. Jacques, Steven L. *In Optical-Thermal Responses of Laser-Irradiated Tissue, 2nd Edition.* s.l. : Springer, 2009.

3. *Kahn, Herman. Use of different Monte Carlo sampling techniques. Rand Corporation, 1955.*

4. *Removal of impalpable contraceptive implants.* M Thirukumara, K E Karunakaranb, R Sureshinic. 2019, Sri Lanka Journal of Obstetrics and Gynaecology, p. 51.

5. *In Vivo Measurement of Optical Properties of Human Skin.* Takahiro Kono, Jun Yamada. 2019, International Journal of Thermophysics , pp. 10,11.

6. *Optical properties of biological tissues: a review.* Jacques, Steven L. 2013, IOP Science , pp. 8,15.

7. *Determination of optical properties of human tissues obtained from parotidectomy in the spectral range of 250 to 800nm.* Eric L. Wisotzky, Florian C. Uecker, Steffen Dommerich, Anna Hilsmann, Peter Eisert, Philipp Arens. 2019, journal of biomedical optics, pp. 4,5.

8. *Development of a technique to measure local scattering in turbid media.* Shimizu, Shiyang Liang and Koichi. tokyo : Japanese Journal of Applied, 2021.

Appendix

5.1 Software used:

1- Fortran compiler
2- C compiler
3- Gnuplot
4- Inkscape

5.2 The code

The next section is the fortran code used to implement the simulation of a square beam passing through an arm, as well as a gnuplot code for plotting the data, the main source (2) contains a similar code written in c language, but without boundary conditions.

```
program project
implicit none

integer:: Nbins, N
double precision, parameter:: pi=4*atan(1.0d0),c0=2.997925d10,cos90=1.0d-6
double precision:: threshold, chance,Pstatus
 !! logicals
 integer:: Alive=1, dead=0


double precision:: x,y,z, ux,uy,uz, uxx,uyy,uzz,rp,Rarm,rb,rmuscle,Rfat !!
photon's trajectory and position
double precision:: s,psi !! step size randomnum
double precision :: costheta, sintheta, cospsi, sinpsi !! sin fimctions

integer:: Iphoton
double precision:: ifoton, w , absorb

double precision, allocatable:: Csph(:), Ccyl(:), Cplan(:) !!photon
concentration
double precision:: fSph, fcyl, fplan
double precision:: mua, mus,g,mut,nt,albedo !! optical properties
integer:: Nphotons, Nr
double precision:: r_size, r,dr
```

```
integer:: ir
 double precision:: shellV, CNT
  !! dummy variables
  double precision:: rnd ,u ,temp,a
  integer:: i,j
  real:: start, finish

  call cpu_time(start)

  !! input
  mua = 0.0
mus = 0.0
g = 0.90d0
nt = 1.33d0
Nphotons = 10000
r_size = 6.0
Nbins=500
Nr=Nbins
chance=0.1d0
threshold=0.01d0

!! arm radius
 Rarm=4.52
 Rfat=4.384
 Rmuscle=4.126
dr=r_size/Nbins
albedo= mus/(mus+mua)
allocate(Csph(0:Nr), Ccyl(0:Nr), Cplan(0:nr))

!! initialization
Csph(:) = 0
Ccyl(:) = 0
Cplan(:) = 0


r=0

i=1

do while(i<Nphotons) !! loop for all photons


if (mod(i,Nphotons/10).eq.0) then !! progress indicator
write(0,*) i*100/Nphotons,'% done'
end if
 W = 1.0d0
pstatus = ALIVE
!! beam's initialization
x = pi/2*sqrt(-log(rand()))
y = 0
z = 0.01d0

costheta = 2.0d0*Rand() - 1.0d0
sintheta = sqrt(1.0d0 - costheta**2.0d0)
psi = 2.0d0*PI*Rand()
ux = 0.0d0
uy = 0.0d0
```

```
uz = 1.0d0
do while(pstatus==Alive)

rp=sqrt(y**2+(z-Rarm)**2)  !! the position of the photon with respect to the
center of the arm
rb=sqrt((z-1.18)**2+(y-5.65)**2) !! position of the photon with respect to
the center of the bone
if (rp>Rarm) then
albedo=0 !! outside the tissue
else
if (rp<Rarm) then  !! inside the skin
mus=268.01
mua= 11.0
end if
if (rp<rfat) then !! inside the fat tissue
 mus=157.03
 mua=0.15
end if
if (rp<rmuscle) then !! inside muscle tissue
 mus=255.41
 mua=0.2
end if

if (z>0.7  .and. z<1.79) then !! in the blood section

if (y>0.25 .and. y<0.75) then
 mus=188.3
 mua= 11.0
end if
end if

if (rb < 1.15) then !! in the bones
 mus=193.4
 mua=20.0
end if
albedo= mus/(mus+mua) !! calculating the albedo
end if




!! hop
a=rand()
s = -log(a)/(mua + mus)
x =x+ s * ux
y =y+ s * uy
z =z+ s * uz

absorb = W*(1.0d0 - albedo) !! drop
w= w-absorb

r = sqrt(x**2.0d0 + y**2.0d0 + z**2.0d0)
ir = ceiling(r/dr)

Csph(ir) =Csph(ir)+ absorb/mua

r = sqrt(x**2.0d0 + y**2.0d0)
```

```
ir = ceiling(r/dr)
if (ir >= NR) then
ir = NR
endif
Ccyl(ir) =Ccyl(ir)+ absorb

r = abs(z)
ir = ceiling(r/dr)
if (ir >= NR) then
ir =Nr
endif
Cplan(ir)=Cplan(ir) + absorb



temp = (1.0d0 - g**2.0d0)/(1.0d0 - g + 2*g*rand())
costheta = (1.0d0 + g**2.0d0 - temp**2.0d0)/(2.0d0*g)

sintheta=sqrt(1.0d0-costheta**2.0d0)
psi = 2.0*PI*Rand()
cospsi = cos(psi)

if (psi < PI) then
sinpsi = sqrt(1.0d0 - cospsi**2.0d0)
else
sinpsi = -sqrt(1.0d0 - cospsi**2.0d0)
endif


if (1.0d0 - abs(uz) <= 1.0d-12) then !! spin
uxx = sintheta * cospsi
uyy = sintheta * sinpsi
uzz = costheta *sign(1.0d0,2.0d0*uz-1.0d0)

else
temp = sqrt(1.0 - uz * uz)
uxx = sintheta * (ux * uz * cospsi - uy * sinpsi) / temp + ux * costheta
uyy = sintheta * (uy * uz * cospsi + ux * sinpsi) / temp + uy * costheta
uzz = -sintheta * cospsi * temp + uz * costheta
end if

ux = uxx
uy = uyy
uz = uzz
!! monte carlo
if (W < THRESHOLD) then
a=rand()
if (a.le.CHANCE) then
w = w/CHANCE

else
pstatus = DEAD

endif
endif
write(19,*) x,y,z
enddo
```

```
i=i+1
 enddo
write(0,*)   '          100% done'


!! the output
open (file='results.txt', unit=11)
do ir=1,Nr
r = (ir + 0.5d0)*dr
shellV = 4.0d0*dr*PI*r**2.0d0
Fsph = Csph(ir)/(Nphotons*shellv)

write(11,*) r, fsph,ir
enddo
call cpu_time(finish)

 print*, finish-start

deallocate(Csph, Ccyl, Cplan)



end program
```

5.3 graphing code

next is a code used for graphing the results using gnuplot

```
set term png

set output "graph.png"

set xrange[0:6]

set logscale y 10

plot "{insert file name 1 }" u 1:2 w l lt rgb "#FF0000" title '{insert title 1}', "{insert file name 2}" u 1:2 w l lt rgb "#00FF00" title '{insert title 2}',"{insert file name 3}" u 1:2 w l lt rgb "#0000FF" title '{insert title 3}'
```

Acknowledgements

I would like to thank my supervisor for all the support and the effort he gave me. I am really grateful for all the experience I got from him. I learned a lot form his knowledge. He always motivated us to learn more in science and do the best that we can.